

AYUSH DWIVEDI

Software Development Engineer (SDE) | Python & Backend Developer | Data Structures & Algorithms

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PROFESSIONAL SUMMARY

B.Tech Computer Science student with a strong foundation in Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, and Computer Networks. Experienced in designing and building backend systems, REST APIs, and full-stack web applications using Python, JavaScript, React.js, and Node.js, writing efficient, scalable, and well-documented code. Hands-on with Git-based version control, debugging, and automation scripting. CEH Certified with working knowledge of secure coding practices and Linux systems. Seeking Software Development Engineer (SDE), Backend Engineer, or Full-Stack Developer internship opportunities.

TECHNICAL SKILLS

Core CS Subjects: Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, DBMS, Computer Networks, Time & Space Complexity Analysis

Programming Languages: Python, C++, SQL, Bash, JavaScript (basics)

Backend & Web Development: REST API Design, Node.js, Express.js, React.js, MongoDB, JWT Authentication, HTML, CSS

Tools & Platforms: Git, GitHub, Postman, Linux (Ubuntu/Kali), VS Code

Practices: Problem Solving (LeetCode, CodeChef), Debugging, Code Optimization, Agile/SDLC, Version Control

Security (Supporting): Network Security, Ethical Hacking, OWASP Top 10, Nmap, Burp Suite Basics

INTERNSHIP EXPERIENCE

Full-Stack Developer Intern | Webstack Academy (Remote)

Mar 2026 – Apr 2026

- Developed responsive full-stack web applications using React.js, Node.js, and MongoDB with REST API integration and JWT authentication, collaborating across the full agile SDLC.

Python Developer Intern | Cognifyz Technologies (Remote)

Jan 2026 – Feb 2026

- Built Python-based automation scripts and backend utilities for data-processing tasks using Git-based version control, applying debugging and code-optimization best practices.

IT Intern | Honda Cars India Limited, Greater Noida

Jun 2025 – Jul 2025

- Supported enterprise-level IT operations including system performance monitoring, anomaly detection, and infrastructure maintenance across large-scale environments.

IT Intern | Eduversity (Remote)

Oct 2023 – Nov 2023

- Worked on networking and system-related tasks in simulated Linux environments, gaining practical exposure to Linux systems and network fundamentals.

PROJECTS

EthHack AI — AI-Powered Penetration Testing Assistant — Node.js, Express.js, OpenAI API, MongoDB

- Designed and built a full-stack platform integrating OpenAI APIs for workflow automation; implemented JWT authentication, bcrypt password hashing, Helmet.js, and API rate limiting in a modular backend architecture.

Sudoku Solver — Python, Algorithms, Data Structures

- Built a Sudoku-solving engine using backtracking and recursion to solve constraint-satisfaction problems, optimized time complexity for faster puzzle computation.

AI Chatbot — Python, OOP, System Design

- Developed an AI-based chatbot with a modular, scalable, object-oriented architecture, cleanly separating conversational logic from dynamic response handling.

MedSecure — Secure Medical Data Transmission System — Python, Cryptography, Algorithms

- Engineered a secure data-transmission system using AES/ECC encryption and LSB steganography algorithms for end-to-end confidentiality, meeting healthcare data security standards.

EDUCATION

B. Tech in Computer Science Engineering, Bennett University, Greater Noida

2023 – 2027

CGPA: 7.38/10 (through 5th Semester)

Escorts World School, Kanpur — Class XII (PCM): 83.7% | Class X: 87.3%

CERTIFICATIONS

Certified Ethical Hacker (CEH), EC-Council (Feb 2026) • Ethical Hacker, Cisco Networking Academy (Dec 2025) • Introduction to Cybersecurity, Cisco Networking Academy (Nov 2025) • Databases and SQL for Data Science with Python, Coursera (Oct 2024) • Computer Networks and Network Security, Coursera (Mar 2025) • Operating Systems and You, Coursera (Mar 2025) • Object-Oriented Data Structures in C++, Coursera (Oct 2024)

ACHIEVEMENTS & EXTRACURRICULAR ACTIVITIES

2nd Position, IEEE BU Scavenger Hunt (2024) • Participant, Microsoft Innovate Hackathon (2025) • Deloitte Australia Cyber Job Simulation (Dec 2025) • Active member, FOSS BU Linux & open-source community; attended Linux Fest and Re-Decrypt events (2025)